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Visualization of Human Cognitive States Monitored by High-density EEG Arrays

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Abstract

There is an increasing interest in monitoring mental activities associated with different cognitive states using electroencephalography (EEG). In recent work, we introduced a comparative study of EEG recordings with participants in various cognitive modalities, including open eyes, open eyes with visual stimuli, closed eyes, and meditation. We showed initial evidence that external sensory input stimulate neural activity in a range of frequency bands, while self-induced peaceful states via meditation or relaxation with closed eyes, tend to produce activity focused in the mid-range (alpha band). We applied our movie making methodology to qualitatively explore brain dynamics using the art of encephalography in a novel way. In the present study we test the technology using a 256 electrodes dense-array EEG developed by EGI. We studied cognitive modalities: meditation, closed eyes, math problem solving, and open eyes. The applied temporal and spatial Power Spectrum Density analysis is able to discriminate between these modalities.

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Keywords: Cognition; EEG; Analytic Amplitude; Analytic Phase; Hilbert Transform; Consciousness; Meditation; Emotions; Awareness; Intentionality; Spiritual Values

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1. Introduction

Synchronization and de-synchronization between different oscillatory systems in the body, like the heart, the brain, the respiratory system and others, play an important role in describing different cognitive states with different emotional content, as well as their associations to different states of relaxation or stress. In order to understand this complexity, a sound methodology is required with the use of both linear and non-linear dynamics, aiming at explaining intentional action and decision-making, and particularly concerning cognitive states which are associated with meditative states and inner peace [1-6]. Sensory input stimulates neural activity in various frequency bands [7-9]. For example, auditory input from verbal stimuli in guided meditations or arithmetic operations, that have been reported to produce Theta, Beta and Gamma activity in the temporal area amongst others [5-6]. We have made some steps towards consolidating a methodology [10-12], and set of tools that allow for a better exploration and understanding of meditative states in contrast with more energy demanding states, like the processing of arithmetic operations after verbal request. The tools will support the analysis of brain dynamics in different cognitive states, together with the use of a heart rate variability (HRV) monitor [13-15].

This paper will focus on three (3) modalities, where the brain activity of three (3) adult volunteers was measured in the modalities of Meditation (MED), MathMind (MM), and Open Eyes (OE). Experiments took place in a large meeting room and were conducted using a HydroCel Geodesic Sensor Net (HCGSN), 256 electrodes dense-array electroencephalography (EEG) produced by Electrical Geodesics, Inc. [16-17]. The main goal of this study is to apply signal-processing techniques and a movie making methodology developed previously [7-10], to distinguish and differentiate between various brain states associated to the different conditions or modalities (MED, MM and OE). This we conjecture will give us further knowledge and understanding of neural correlates of consciousness (NCC) [18-21, 24-25], based on a technology that covers the whole scalp and facial areas with a significantly improved spatial resolution.

2. Experimental Equipment and Methods

The EEG technology used for this study was a HydroCel Geodesic Sensor Net (HCGSN), 256 electrodes dense-array electroencephalography (EEG) produced by Electrical Geodesics, Inc. [17]. This company produces HCGSN in two different configurations (with and without sponge inserts), in a range of sizes and varying densities (32, 64, 128, or 256 channels). In this study, we used an adult medium sized (56–58 cm) 256-channel HCGSN with sponge inserts. The design and structure of the HCGSN is based on a geodesic tessellation of the head (see Figure 1, Left) whereby "each line between sensor pairs is a geodesic, the shortest distance between two points on the surface of a sphere." The Geodesic Sensor Net (GSN) provides millimeter accuracy in positioning sensors, quick application time and a comfortable net well adapted to the head shape amongst other advantages. In the EEG equipment, the sensors measure changes in voltage at the surface of the scalp and they also pick up electrical noise from the environment. These EEG signals are received by amplifiers (simultaneously) and then, they are amplified, filtered, sampled, and digitized. After that, they are sent from the amplifier to the data-acquisition computer via a Universal Serial Bus (USB) cable or a fire wire cable (Net Amps 300).

Figure 1 (Right) shows the EEG sensor net channel map [†] containing the 256 electrodes according EGI specifications. Next, the measured channels were remapped using a 16x16 matrix corresponding to the 256 EEG electrodes, as shown in Figure 2 (Left). This mapping was a challenging task and we had to exclude some of the 256 channels in order to adapt the square 16x16 representation to the shape of the brain. Occasionally we duplicated an electrode, just next to itself (in bold), in order to preserve the symmetry of the brain in relationship to the EEG sensor net and the matrix. We also left some electrodes in zero (0) both for representation purposes and as markers to facilitate recognition of the different areas of the head in the brain dynamics movies and figures.

[†] Our appreciations are due to the team of EGI, via email correspondence with Katia Krane, for permitting us to include the channel map in this paper in order for us to create the color codes on it to facilitate the making of the matrix positions, as shown in Figure 1 (Right).

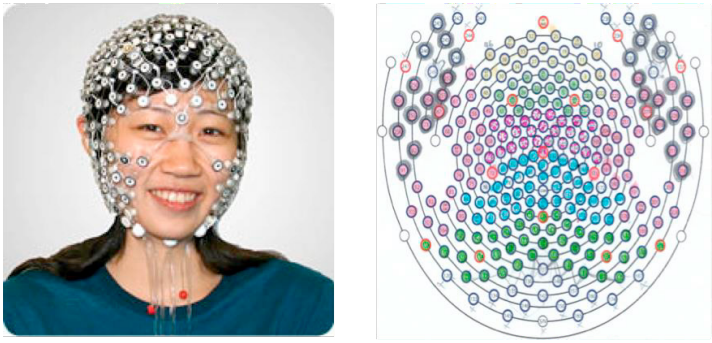

Fig. 1. Left: EGI HCGSN 256 electrode dense-array EEG [17]. Right: Sensor Net 256-Channel Map that covers the following areas of the brain: (1) Prefrontal (Fp, yellow), (2) Frontal (F, olive green), (3) Temporal (T, pink), (4) Central (C, magenta), (5) Parietal (P, turquoise), (6) Occipital (O, green), (7) Facial (F, grey).

Note that matrix (A) reads as any ordinary array in linear algebra where A(i,j) represents an element of row 'i' and column 'j'. For example, A(1,1)=246 meaning it is associated with electrode 246 of the EEG sensor net while A(2,13)=1 is associated with electrode 1 of the EEG sensor net. Finally, we should note that this 16x16 matrix (A), was also represented as a column vector (v) 256x1 for computational purposes where A(1,1) = v(1), A(1,2) = v(2), … , A(1,16) = v(16), A(2,1) = v(17), ... , A(16,16) = v(256). From now on whenever we refer to channels we will name the electrode number as shown in the sensor net, while when we refer to the position in the matrix, we will use a sequential number between (1 to 256), as described above by the relationship A(i,j)=v(k) $\forall$ i,j=1,16 and k=1,256. For example, the associated channels for the front left side are to be found in positions 1, 2, 3, 4, 17, 18, 19, 33, 34, 49, 50, 51 and 52 for channels 246, 243, 245, 248, 254, 253, 252, 73, 67, 255, 250 251 and 82 respectively. For the front right side we have to identify them by symmetry in the matrix. This will allow us to relate to the different areas of the brain with ease while looking at figures and watching brain dynamics movies.

The experiments were performed by a research group of volunteers in support of the Science and The Embassy of Peace (SOP and EOP, respectively), in collaboration with representatives from Electrical Geodesics, Inc. (EGI). While the EGI team prepared the sensor net, applied it to the participants' head and operated the EEG system, the EOP team monitored the participants' heart rate and other physiological measurements, operated the sound system for the experiments as needed, and kept a written log of the experimental procedures. The participants were also connected to a HeartMath emWave Pro device that measured their Heart Rate Variability (HRV) [22-23]. Participants also wore a Sleep Tracker to detect their awake/sleep mode. In addition to the EEG and HRV measurements, participants' pulse and blood pressure were also recorded before commencing the experiments. Experiments were conducted in a large meeting room of a hotel in Auckland, New Zealand, which was heated to a warm temperature (~23°C), had overhead electric lighting (50-60 Hz) and large windows letting in full light from outside. The experiments took place in the middle of the day between 12 PM and 2 PM.

We use three (3) experimental modalities for this study. Each of these three (3) modalities was conducted for duration of 7 minutes, so a full experiment took 21 minutes, with markers indicating a change in modality. For each experiment the participant was positioned in a comfortable chair faced in the direction of light from the windows.

Modality 1 - Meditation (MED) - The objective of this experiment was to measure neural activity in different regions of the brain as the participants practiced their preferred form of meditation (with closed eyes) for duration of seven (7) minutes.

Modality 2 - MathMind (MM) - The objective of this experiment was to measure neural activity in different regions of the brain as the participants listened to a recording of arithmetic exercises (varying in difficulty) and endeavored to find the solution silently in their mind for duration of seven (7) minutes. The arithmetic operation was pre-recorded with a soft voice, usually in a sequence of three (3) simple problems followed by one (1) difficult problem. The time between each question was set to around five (5) seconds for easy equations and fifteen (15) seconds for harder questions, allowing the participant more time to calculate the answer.

Modality 3 - Open Eyes (OE) - The objective of this experiment was to measure neural activity in different regions of the brain as the participant sat with eyes open, blinking minimally, for duration of seven (7) minutes.

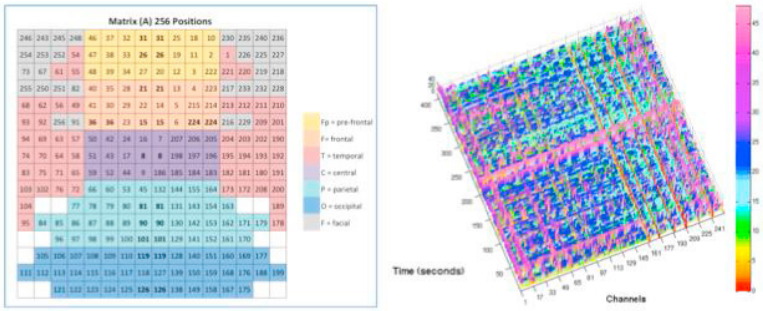

Fig. 2. Left: A representation matrix (A) of 16x16 for the EEG sensor net over the whole scalp. This matrix covers the following areas of the brain: (1) Prefrontal (PF), (2) Frontal (F), (3) Temporal (T), (4) Central (C), (5) Parietal (P), (6) Occipital (O), (7) Facial (F). The left and right hemispheres divide the matrix in two (2) equal and symmetrical parts comprised of two (2) sub matrices of 16x8 each. Right: Dominant Frequency Band (DFB) EEG activity for the 256 channels, for 400s, for participant 2 in the MathMind (MM) modality.

The EEG signal processing methodology used is the same as described in, and also inspired by, previous work [12, 24, 26-27]: (a) importing and preprocessing the data; (b) computing the temporal power spectrum (PSDt) for each of the channels included in the 16x16 matrix, in windows of 1 second; (c) analysis of the PSDt together with quantitative measures based on power and frequency band, over different cognitive modalities. For the frequency bands analyzed; see Table 1.

Table 1. Frequency windows analyzed.

Frequency band	Windows (Hz) $[f_{Low} - f_{High}]$	Band power sum over $[f_{Low} - f_{High}]$
Theta	4-6	TP
Alpha	8-12	AP
Low Beta	14-18	LBP
High Beta	20-26	HBP
Low Gamma	28-36	LGP
High Gamma	38-48	HGP

Following [12], we compute the dominant frequency band (Pband), based on the PSDt of each band as follows:

$$P^* = max\{TP, AP, LBP, HBP, LGP, HGP\} \quad \text{Maximum Power} \tag{1}$$

$$DFB = fBand(P*) \quad \text{Dominant Frequency Band} \tag{2}$$

Note that P* stands for the maximum power to be selected from the set of the different frequency band powers (TP, AP, LBP, HBP, LGP, HGP), where DFB results as the Dominant Frequency Band in terms of power represented as fBand(P*). “fBand” is the frequency band associated with P*. Based on the dominant frequency band (DFB) for each window of 1 second duration and for each channel in the 16x16 matrix, we produce spatio-temporal movies. Also, we create a set of figures where the DFB is displayed for all 1s windows for all channels in the 16x16 matrix. Finally, we perform a statistical comparative analysis between modalities. Beyond the temporal analysis (PSDt), we may be able to compute the spatial power spectrums (PSDx) based on the spatial resolution that is

provided with this EEG technology, following previous work [11]. However, such spatial analysis is beyond the scope of this study.

3. Experimental Results

First we introduce the basic display formats to represent the evolution of the DFB per channel (256 channels) along the time axis. From the original 21 minutes recorded for each participant, encompassing three (3) modalities (MED, MM and OE) of 7 minutes each, we only studied a period of 400 s (~6.66 min) per modality. This was done in order to remove the initial startup period for the first part of the data and to then analyze equally sized segments for each modality.

Figure 2 (Right) illustrates several main observations using the example of MM modality. For example, positions 1 to 4, and 13 to 16, (for vector v from the 16x16 matrix) display predominately Gamma frequency for most of the 400 s period. These positions are associated to the facial left and right areas respectively. Positions 5 to 12 are mainly dominated by Alpha and Beta and are associated with the very base of the prefrontal area; see Figure 1 (Right) and Figure 2 (Left). About half way through the experiment (~220 - 225 s) most of the electrodes show High Gamma activity in a form of large synchronization, while the whole brain shows more Alpha activity in the second half. The markers or positions with a value of zero can be clearly seen between positions 161 and 241. Overall, the back of the brain (occipital and parietal regions) seems to be dominated by Alpha with some Beta while the Frontal and Central areas are more dominated by Gamma and High Beta. The brain is highly dynamic in relationship to the shifts that it displays from frequency band to frequency band, something that can be observed with even more clarity by watching the brain dynamics movies.

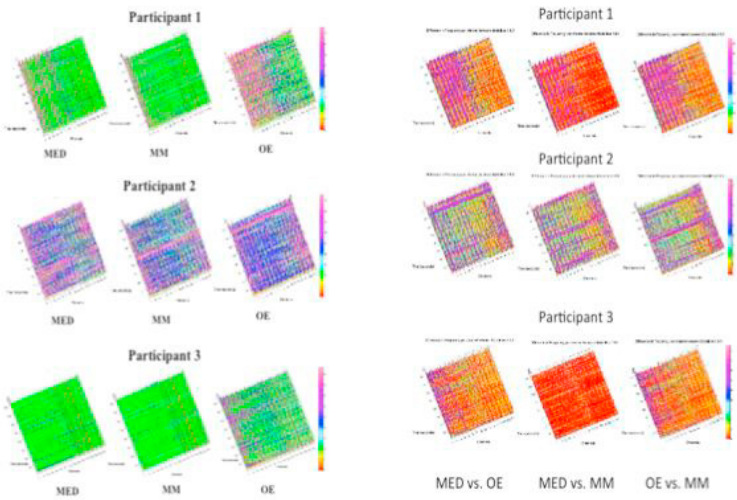

Fig. 3. Left: Shows the 256 positions associated with different channels for the EEG Dominant Frequency Band (DFB) for 400s for the three (3) participants in all modalities. Top: participant 1, Middle: participant 2 and Bottom: participant 3 with MED, MM, OE, left to right for each participant. Right: Shows a comparison between modalities: (1) MED vs. OE, (2) MED vs. MM and (3) OE vs. MM for: participant 1 (top), participant 2 (middle) and participant 3 (bottom).

Figure 3 (Left) gives an overview of the Dominant Band behavior across modalities for the three (3) participants. We can clearly observe both the differences between modalities as well as between participants. Such individual differences are expected since every participant's brain is unique in configuration and the way it processes information about different cognitive states [12]. Also, it is relevant to mention that participant 1 and 2 are the same as in [11]. Now we have added participant 3 with the knowledge a priori that participant 1 and 3 are advanced meditators, while participant 2 has started practicing meditation regularly in more recent times. In accordance with

[12], participant 2 shows more brain activity in Beta and Gamma, while participants 1 and also 3 display significantly more Alpha frequency most of the time, in most brain areas across all modalities.

The most relevant differences between modalities per participant when contrasting them in terms of the dominant frequency band (DFB) are:

- **Participant 1**: MED shows Alpha most of the time for most channels with some Low Beta and Low Gamma activity sparsely distributed for most channels most of the time. However, we clearly notice the presence of Gamma activity, both low and high, in the temporal areas, as well as some of the facial area, which is also expected to capture temporal activity. MM shows similar results to MED, however with more Alpha activity present for most channels. OE shows a quite remarkable difference between the temporal, prefrontal and frontal areas with abundant High Gamma in contrast with the central, parietal and occipital lobes, where we observe predominantly Alpha activity.
- **Participant 2**: We observe very similar brain dynamics for the three (3) modalities (MED, MM and OE) with abundant Beta and Gamma and significantly less Alpha when contrasted with the other two (2) participants.
- **Participant 3**: We see a very similar kind of activity to participant 1, the only noticeable difference, so far, is observed in OE where this participant shows less activity in Gamma and more Activity in Alpha for most of the brain areas most of the time.

After the above preliminary analysis we endeavored to explore with great care what the spatio-temporal brain dynamics showed us in the movies we created, based on the representation matrix of the brain that we produced. We certainly observed remarkable differences between participants and modalities, however we also observed some similarities, veiled to us in the previous analysis, concerning the activation and deactivation of certain areas of the brain when they dynamically change from Alpha to Theta, Beta or Gamma when performing arithmetical operations or with Open Eyes in contrast to Meditative states. Next, we present some examples of different participants and modalities for only four (4) frames of duration of 1 s.

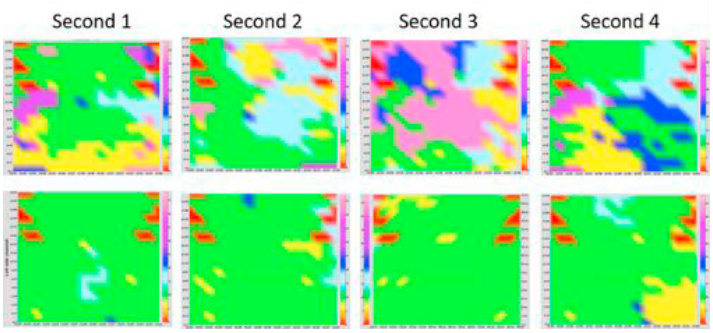

Fig. 4. EEG Dominant Frequency Band (DFB) plot for the 16x16 representation matrix during 4 consecutive seconds: Top: participant 2; Bottom: participant 3, for the MED modality.

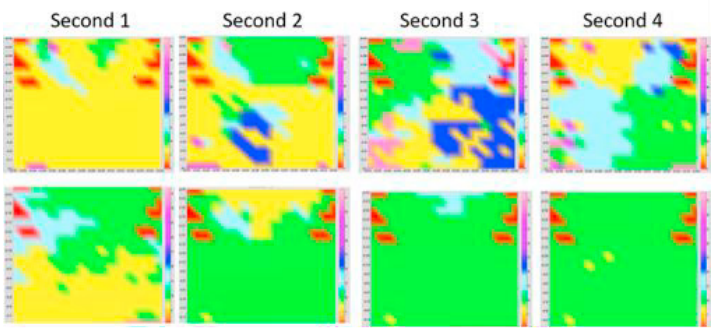

Fig. 5. EEG Dominant Frequency Band (DFB) plot for the 16x16 representation matrix during 4 consecutive seconds: Top: participant 2; Bottom: participant 3, for the MM modality.

In Figures 4 and 5, we observe differences between modalities MED and MM for the four (4) frames displayed. Generally speaking, participant 2 shows a diverse range of frequencies in brain dynamics most of the time for all modalities. However, in the MED modality we can see intermittent changes between Alpha and other bands like Theta, Beta and Gamma, with more Alpha present than in the MM modality. For the modality MM, we saw shifts between Theta and Alpha for both participants, somehow associated with the arithmetic processing times and the degree of difficulty, something which appears to trigger different dynamics between different areas of the brain, shifting to Theta, Beta or Gamma and seemingly returning to Alpha. Again, this requires more movie watching and further research, which is outside the scope of this work.

It is interesting to mention that many of the brain dynamics patterns that we observe in the movies, we have already observed in previous work [12], like rotational and pulsation patterns showing the areas of the brain that are more active in Alpha, Theta, Beta or Gamma for the different modalities. So far for us, this seems to be related to the nature of the cognitive states associated with the modalities, as well as the particular level of meditative mastery of each participant in the context of their particular brain processes based on genetics, environmental conditions, learning and habitual differences. We need to stress here the importance of watching the movies in order to capture the patterns associated to each modality's brain dynamics. This clearly requires more research, though so far it looks very promising to us.

In Figure 3 (Right), we show a comparative set of graphs for the three (3) participants, where we can observe the differences between modalities for each of them. In order to produce these graphs we calculated the difference between the 16x16 matrices associated to each modality for each display which shows that, generally speaking, for all participants we observe a larger difference between the frontal and temporal lobes than the central, parietal and occipital ones, except for modalities MED and MM that show very small differences across the whole scalp for participants 1 and 3. Participant 2 shows larger differences between modalities, which are very similar for all three (3) comparative plots. Participant 1 shows significantly more difference in temporal and frontal activity than participants 2 and 3 for all three (3) comparative plots.

Next, we conducted an initial statistical analysis of both: (a) the mean frequency (see Eq. 3) and (b) the mean of the differences in frequency between modalities (see Eqs. 4-5) and we collapsed the MFAC(t) and the MDIFF(t) data into Boxplots. The methodology we used for this purpose was the same as in [12], where we calculated both the MFAC(t) and the MDIFF(t) for each 1s window for all the channels in the 16x16 matrix (256 positions). We did this for every modality and participant using the subscript "c" to indicate the modality for the MFAC(t). The Boxplot analysis was also conducted on both sets of data (modalities and participants) as shown in Figure 6 (Top).

$$MFA(t)_c = \sum_{c=1}^{256} \frac{DFB(t)_c}{256} \quad ; \quad \forall \ t = 1,2, \dots 400 \tag{3}$$

We observe in Figure 6 (Top) that on one hand both participant 1 and 3 show similar tendencies and median values for MED and MM while for OE they both show higher median values and greater spread, however, participant 3 shows smaller values and smaller spread than participant 1. On the other hand, participant 2 shows very similar values and spread for the three (3) modalities and in general those values are significantly larger than the values for participant 1 and 3. At this stage, it is important to remember that participant 1 is more advanced in meditative states than participant 2, and both participant 1 and 3 are more mature or advanced than participant 2. Therefore it is expected that their MFAC(t) probability distributions will differ for most modalities. In Table 2 we display the approximate median values for each participant in each modality.

Table 2. MFAC Medians for all Participants & Modalities

Modality	Participant 1 Median (Hz)	Participant 2 Median (Hz)	Participant 3 Median (Hz)
MED	~12	~15	~10
OE	~14	~15	~12
MM	~11	~14	~10

Here we present the second type of analysis that we conducted, this time based on the differences between modalities. Equations 4 and 5 show the set of first computations.

$$DIFF_c = abs\{DFB(t)_{jc} - DFB(t)_{kc}\} \quad (4)$$

For $k \neq j$ and $\forall\ c = 1,2, \ldots 256$, where k and j are the different modalities MED, OE, MM and therefore, k = 1,3 & j = 1,3. Note that when k=1 and j=3, for example, that DIFFc is the same as when k=3 and j=1, and therefore it is redundant since the values would be the same. Once we have done the first set of computations, we apply to DIFF(t)c a similar treatment as in Eq. 3 and we obtain the mean of the differences in frequencies, to prepare a new set of Boxplots, now with the MDIFF(t) vector per modality for each participant.

$$MDIFF(t)_c = \sum_{c=1}^{256} \frac{DIFF(t)_c}{256} \ ; \ \forall\ t = 1, 2, \ldots\ldots 400 \quad (5)$$

where $MDIFF(t)_c$ = MDIFF(t) for each modality c. The contrasted modalities are as follows: MED vs. OE, MED vs. MM and OE vs. MM with their corresponding sub-indices k, and j, as used in Eq. 4.

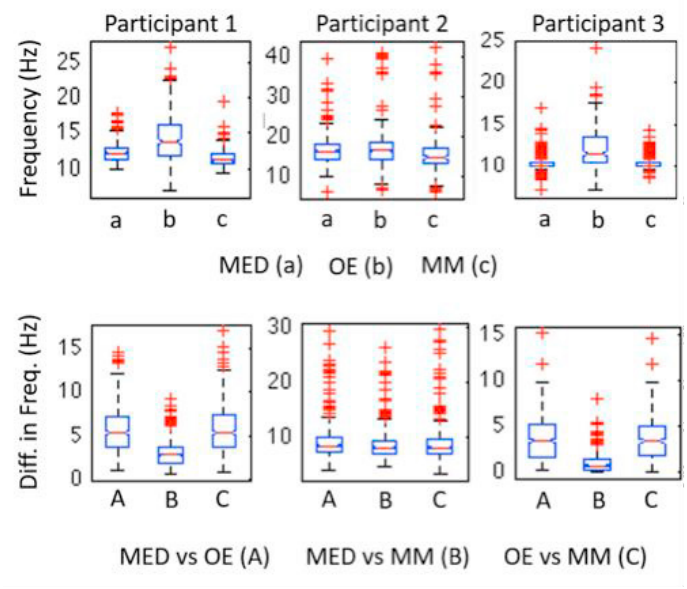

Fig. 6. Top: Boxplot analysis for the vectors of the Mean Frequency for all channels calculated for every 1s window, $MFA(t)$, for all modalities, for each participant. Bottom: Boxplot analysis for the vectors of the Mean Frequency of the differences between modalities for all channels, calculated for every 1s window, $MDIF(t)$, for each participant.

In Figure 6 (Bottom) we display the Boxplots of the mean differences MDIFF(t) between modalities and we summarize our findings as follows:

- OE presents very significant median differences from both MED and MM, with large spreads for both participants 1 and 3.
- Participant 2 shows very similar median values for all differences between modalities and with similar spreads and outliers.
- Both participants 1 and 3 show similar small median values with small spreads for MED vs. MM differences in frequency. This indicates that somehow MM is very similar in brain dynamics to MED, particularly for participant 3.
- Apart from MED and MM that are quite similar in brain dynamics for the three (3) participants, the other modality (OE) is quite different to both MED and MM, showing a large spread when compared with both of them (MED and MM) for participants 1 and 3 only.
- We conclude with computing the norm or Euclidean distance (ED) of the MDIFF(t) vector, which gives a measure for the average distance between modalities, as shown in Figure 7 (Left and Right), together with the confidence intervals for the mean of DIFF(t) and 95% confidence intervals. The EDs show very clearly what we have already observed in our analysis so far, however, now we start to have a taste for the statistical significance between the modalities contrasted.

It is clear from both the EDs in Figure 7 (Left and Right), that the difference between MED vs. MM is significantly smaller even for participant 2 who is a less masterful meditator. Moreover, participant 3 shows significantly more Alpha activity for all modalities than participants 1 and 2. The fact that the OE modality shows the greater MDIFF(t) between both MED and MM modalities for all participants is also relevant.

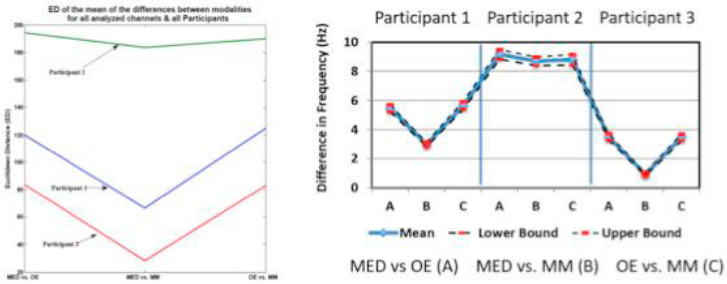

Fig. 7. Left: Euclidean Distance (ED) for the vectors of the Mean Frequency of the differences between modalities for all channels, MDIFF(t), calculated for every 1s window, for each participant, where Participant 1 (Blue), Participant 2 (Green) and Participant 3 (Red). Right: Shows mean and confidence intervals with □=0.05 for the vectors of the Mean Frequency of the differences between modalities for all channels, $MDIF(t)$, calculated for every 1s window, for each participant.

4. Discussion

We have used a diverse set of tools and algorithms to understand brain dynamics both qualitatively and quantitatively, following [12]. We have derived valuable information from this analysis concerning the transitions between dominant frequency bands that were recorded from the 256 electrodes EEG sensor net. Both spatial and temporal patterns are regarded as one (1) object when we look at the movies to identify patterns associated to each modality, various stimuli (mainly auditory and visual) and participant. Some of the relevant observations made via the art of electro-encephalography via brain dynamics movies are:

- Modality MED is significantly dominated by Alpha activity for participants one (1) and three (3) and shows more Alpha activity also for participant two (2) though with abrupt transitions between Theta, Beta and Gamma. It seems that this participant is more mentally active in her imagination, more likely entertaining visual images and mental chatter or perhaps very self-aware or aware of auditory stimuli from the environment. This is reflected in Beta activity in the occipital lobe, as well as high Gamma for the frontal and temporal lobes, typical of humans engaged in mental energy-consuming activities, as in the creation of knowledge and meaning [24-25].
- Modality OE shows an increasing number of transitions between the different dominant frequency bands and lobes of the brain for all participants. It is clear from the movies that the occipital lobe is still dominated by Alpha activity with occasional transitions to Beta and Gamma for participants one (1) and three (3), and for the prefrontal, frontal cortex we observe alternations between Theta and Gamma, while the temporal lobe is mainly dominated by Gamma with some Beta activity. These dynamics may have arisen as a consequence of an intense visual stimulus while staring at the white wall with open eyes, while listening to some occasional and perhaps interesting background conversations with soft voices.
- Modality MM acted as a kind of Vipassana guided meditation [23, 28], however with the difference that the auditory stimuli was a set of questions with a request to perform simple mental arithmetic computations. These requests were made by a cd audio that allowed a space of ~ 5s between simple questions and ~ 15s for the difficult questions. Here we could detect many transitions between Alpha and Theta in the prefrontal and frontal areas, together with Gamma in the temporal lobes.

In Figure 8, we show the counts of number of seconds per electrode in the 16x16 representation matrix per band, together with the bar charts (far right) of the number of seconds spent per band. This display supports us to begin to unravel the role played by each lobe in each modality. For example, for participant 3, we observe that the brain spends most of the time in Alpha both for MED and MM. However, the fronto-temporal-facial and in particular occipital and parietal show activity in L-Beta, predominantly in the left hemisphere. The Theta band is also active in

the pre-frontal, frontal and temporal lobes. The central and parietal lobes show mild activity in the H-Beta band while the Gamma band is very quiet with some activity in the temporal lobes.

For the OE modality, as we have mentioned before, the brain dynamics are quite different. For example, there is significantly less Alpha present in the experimental period, while Theta, Low Beta and even H-Beta show a very noticeable increase in activity, particularly in Theta for most parts of the brain, predominantly higher in the pre frontal and frontal, as well as lower activity in the occipital-left area. Also, L-Beta becomes very present for most parts of the brain, where the frontal area shows mild activity, the central and parietal areas show medium activity and the occipital area shows very strong activity.

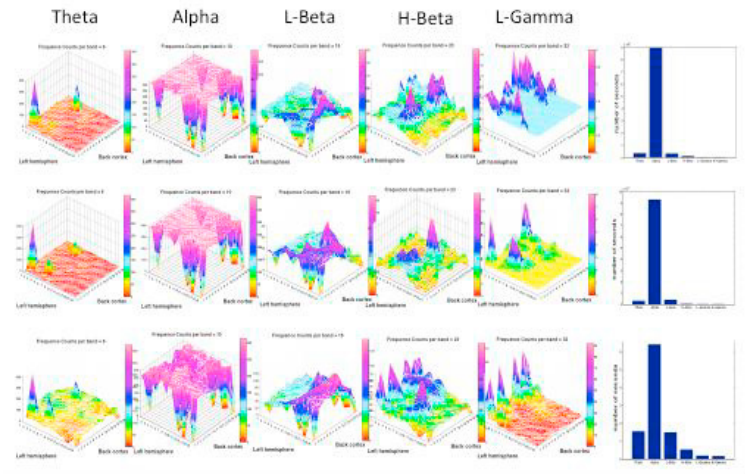

Fig. 8. Display of the total of seconds spent by the brain in the different bands for each electrode included in the 16x16 representation matrix for the Theta (6 Hz), Alpha (10 Hz), L-Beta (16 Hz), H-Beta (23 Hz), L-Gamma (32 Hz). Top: MED, Middle: MM and Bottom: OE. Last column displays the total of seconds spent by the brain per band. All displays are for participant 3.

The changes between modalities in terms of brain activity can be explained when we take into consideration that, for example, in MED the participant is in a relaxed mode with her eyes closed with no or very little visual stimulation (photons penetrating the skin) and with very gentle and soft background conversations that occasionally occurred. This should, for an experienced meditator, show predominantly Alpha activity with none or very little occipital activity (due to visual stimuli) and the same applies for the temporal lobes (due to mild auditory stimulation). However, the OE activity is a very visually demanding one, since the participants are asked to stare for ~7 minutes at a white wall and to blink as little as possible to minimize artifacts. This, as it does, should show significantly more activity in the occipital lobe (due to intense visual stimuli). Since participant three (3) is the most advanced meditator, we conjecture that she is able to regularly enter Alpha states even in the face of visual and auditory stimuli and this transition we also conjecture is in close relationship with the presence of Theta and Beta activity.

We applied this methodology successfully one more time and feel that we have gained additional understanding of brain dynamics and the benefits of meditation in generating coherent inner states dominated by Alpha and cognitive states that arise by the subjective response to a request to perform an arithmetic operation followed by the use of the mental capacity to objectively and rationally come up with the answer. We have expanded our results from previous work [11, 12, 27] by using a more powerful EEG technology and adding a third reliable meditator as a new participant to improve our understanding of the difference between modalities and participant's brain dynamics with an extra bit of statistical significance. We aim at adding additional participants and modalities to our future research.

5. Conclusions

The work we have presented here is in agreement with our previous study [11-12], where Power Spectrum Density (PSD) can be used to facilitate the process of qualitatively and quantitatively discriminating between modalities and participants. We rely on previous work related to brain dynamics in the creation of knowledge and meaning [10-12, 18-19, 24-27]. Our main conclusions can be summarized as follows:

- We successfully measured three (3) participants in three (3) different modalities and performed a strong analysis in order to compare different cognitive states associated to the three (3) modalities (MED, OE and MM) with their corresponding brain dynamics patterns.
- We applied our methodology, this time with the aid of a powerful EEG technology with 256 electrodes. The movies we produced were significantly better than our previous ones, concerning spatial resolution. This allowed for a better identification of brain areas involved in different kinds of processing and cognitive states.
- We applied statistical analysis (EDs, means and confidence intervals, as well as Boxplots) to quantitatively compare modalities based on Mean Frequency Band per modality, per participant, as well as the mean of the differences in dominant frequency per modality and participants for all channels included in the 16x16 representation matrix for every 1s window.
- We pointed out important differences between meditators and their responses to different stimuli w.r.t. different modalities.
- In accordance with previous studies, the differences in modalities are smaller in the occipital and parietal regions of the scalp for the majority of the participants.

In the future we intend to do an extensive study about brain dynamics in the different modalities based on this database, since this study is out of the scope of this detailed analysis for all participants. We intend to expand this line of research and aim at improving the understanding of brain dynamics and its relationship to mental health, general wellbeing and particularly inner peace, to possibly improve the tendency towards a peaceful society. It is important to mention that, simultaneously to the EEG recordings, we also measured the participant's Heart Rate Variability (HRV) and monitored them with a sleep watch. We found interesting connections between HRV and brain dynamics; however, this analysis is the subject of future studies.

It is fascinating to observe these dynamics and let the imagination loose to gaze at possible scenarios of the different pathways that are activated by different kinds of stimuli, together with internal and subjective states for self-regulation and self-mastery of coherent and peaceful states, at will, via meditation for example. It certainly inspires us to keep going with our challenging exploration. This exploration will require new and more advanced statistical and mathematical analysis with new methods and algorithms to properly describe linear and non-linear brain dynamics.

We feel very comfortable in terms of the minimal amount of artifacts associated with EMG due to both the fact that, the participants are well trained in achieving relaxed body states with minimal body and particularly facial movement [11, 26]. We observed moments of Theta and Gamma whole brain synchronization events, the Gamma one happened between seconds 200-255 for participant 2, for modality MM. This event could presumably be associated to a massive artifact, however, it is from this event onwards that participant 2 showed significantly more Alpha activity than before the event, like an onset of a sort for a new dynamics. We conjecture that the future is full of diverse possibilities in what concerns the use of biofeedback systems for personal training in meditation and self-mastery, as well as for the use of therapeutic and health related purposes.

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References

[1] Freeman, W. J., Rogers, L. J., Holmes, M. D., & Silbergeld, D. L. (2000). Spatial spectral analysis of human electrocorticograms including the alpha and gamma bands. (V. Crunelli, Ed.) *Journal of Neuroscience Methods, 95* (2), 111-121.
[2] Buzsáki, G. (2006). *Rhythms of the Brain.* New York, New York, USA: Oxford University Press.
[3] Davis, J. J. (2009). *The Brain of Melchizedek - A Cognitive Neuroscience Approach to Spirituality (Thesis).* Dunedin, New Zealand: Otago University.
[4] Schwartz, J. M., Stapp, H. P., & Beauregard, M. (2005). Quantum physics in neuroscience and psychology: a neurophysical model of mind-brain interaction. *Philosophical Transactions: Biological Sciences, 360* (1458), 1309–1327.
[5] Kasamatsu, A., & Hirai, T. (1996). An electroencephalographic study on the zen meditation (zazen). *Folia Psychiatrica et Neurologica Japonica, 20* (4), 315-336.
[6] Austin, J. H. (1999). *Zen and the Brain: Toward an Understanding of Meditation and Consciousness.* Cambridge, MA, USA: The MIT Press.
[7] Freeman, W. J., & Quiroga, R. Q. (2013). *Imaging Brain Function with EEG - Advanced Temporal and Spatial Analysis of Electroencephalographic Signals.* New York, NY, USA: Springer.
[8] Kozma, R., W.J. Freeman (2016) *Cognitive Phase Transitions in the Cerebral Cortex,* Springer, Heidelberg.
[9] Freeman, W. J., Holmes, M. D., Burke, B. C., & Vanhatalo, S. (2003). Spatial spectra of scalp EEG and EMG from awake humans. *Clinical Neurophysiology, 114* (6), 1053-1068.
[10] Davis, J. J., Kozma, R., & Freeman, W. J. (2015). The Art of Encephalography to Understand and Discriminate Higher Cognitive Functions Visualizing Big Data on Brain Imaging using Brain Dynamics Movies. *Procedia Computer Science . 53*, pp. 56-63. San Francisco: Elsevier.
[11] Davis, J. J., Kozma, R., Lin, C.-T., & Freeman, W. J. (2016). Spatio-Temporal EEG Pattern Extraction Using High-Density Scalp Arrays. *2016 International Joint Conference on Neural Networks (IJCNN)* (pp. 889-896). Vancouver: IEEE Press.
[12] Davis, J. J., Lin, C.-T., Gillett, G., & Kozma, R. (2017). An Integrative Approach to Analyze EEG Signal and Human Brain Dynamics in Different Cognitive States. *Journal of Artificial Intelligence and Soft Computing Research (JAISCR), 7* (4), 287-299.
[13] Childre, D., & Cryer, B. (2008). *From Chaos To Coherence – The Power To Change Performance* (Revised Edition ed.). USA: HeartMath System.
[14] McCraty, R., Atkinson, M., Tomasino, D., & Bradley, R. T. (2006). *The Coherent Heart: Heart–Brain Interactions, Psychophysiological Coherence, and the Emergence of System-Wide Order.* Boulder Creek, CA, USA: Institute of HeartMath.
[15] Medicore. (2017). *Heart Rate Variability Analysis System - Clinical Information - Version 3.0*. Retrieved June 10, 2017, from Medicore Products SA-3000P: http://medi-core.com/download/HRV_clinical_manual_ver3.0.pdf
[16] Electrical Geodesics, Inc. (2007). *Geodesic Sensor Net Technical Manual.* Eugene, OR, USA.
[17] Electrical Geodesics, Inc. (2018). *Geodesic Sensor Net 256-Channel.* Eugene, OR, USA: https://www.egi.com/images/stories/clinical/images/, accessed 01/27/2018.
[18] Pockett, S., Bold, G. E., & Freeman, W. J. (2009). EEG synchrony during a perceptual-cognitive task: Widespread phase synchrony at all frequencies. *Clinical Neurophysiology, 120*(4), 695-708.
[19] Ruiz, Y., Pockett, S., Freeman, W. J., Gonzalez, E., & Li, G. (2010). A method to study global spatial patterns related to sensory perception in scalp EEG. *Journal of Neuroscience Methods, 191* (1), 110-118.
[20] Kasabov, N., & Capecci, E. (2015). Spiking neural network methodology for modelling, classification and understanding of EEG spatio-temporal data measuring cognitive processes. *Information Sciences, 294*, 565-575.
[21] Pribram, K. (2013). *The Form Within: My Point of View.* Westport, CT: Prospecta Press.
[22] HeartMath Inc. (2015, November 11). *Knowledgebase Article - HeartMath Inner Balance Trainer Coherence Scoring System*. Retrieved June 16, 2017, from HeartMath Knowledge: https://www.heartmath.com/support/knowledgebase/?article=kA180000000XbZmCAK&t=HeartMath+Inner+Balance+Trainer+Coherence+Scoring+System.
[23] Mental Health Daily. (2017). *Types Of Meditation: Extensive List of Techniques*. Retrieved July 30, 2017, from Mental Health Daily - Mental Health Blog: http://mentalhealthdaily.com/2015/03/24/types-of-meditation-extensive-list-of-techniques/
[24] Davis, J. J., Kozma, R., & Freeman, W. J. (2013). Neurophysiological evidence of the cognitive cycle and the emergence of awareness. *IEEE 2013 iCAST*, pp.149-156. Aizu, Japan, IEEE Press.
[25] Davis, J. J., Gillett, G., & Kozma, R. (2015). Revisiting Brentano on Consciousness: Striking Correlations with Electrocorticogram Findings about the Action-Perception Cycle and the Emergence of Knowledge and Meaning. *Mind and Matter, 13* (1), 45-69.
[26] Freeman, W. J., Ramon, C., & Holmes, M. D. (2010). 1-D spatial autocorrelation function of EEG: a sensitive assay for occult EMG. *16th Annual Meeting of the Organization for Human Brain Mapping. Abstract #881.* Barcelona: HBM2010.
[27] Kozma, R., Davis, J. J., & Freeman, W. J. (2012). Synchronized Minima in ECoG Power at Frequencies Between Beta-Gamma Oscillations Disclose Cortical Singularities in Cognition. *Journal of Neuroscience and Neuroengineering, 1* (1), 13-23.
[28] Cahn, B. R., Delorme, A., & Polich, J. (2013). Event-related delta, theta, alpha and gamma correlates to auditory oddball processing during Vipassana meditation. *Social Cognitive and Affective Neuroscience (SCAN), 8* (1), 100-111.